

Category	Parameter	Default Configuration	Optimized Configuration (After Testing)	Notes
Core Sensing	Resolution	8×8	4x4	Affects processing load & active time
	Ranging Frequency (Hz)	1 Hz	1 Hz (objects >300 cm from lift), 6 Hz (objects <300 cm from lift)	Controls detection interval & duty cycle
	Timing Budget / Integration Time	Default (10 ms)	Default (10 ms)	Directly sets active window length
	Operating Mode	Autonomous	Autonomous	Autonomous vs Continuous
	Target Order	STRONGEST	CLOSEST	Affects internal processing strategy
	Sharpeners (%)	Default	N/A	Impacts signal processing intensity
	Ranging Profile	Default	N/A	If modified (Long Range / High Accuracy etc.)
Calibration	VHV Repeat Count	1 (Every Frame)	10 (Once every 10 frames)	Controls calibration frequency
	Offset Calibration	Default	N/A	If custom calibration used
	Crosstalk Compensation	Default	N/A	Environmental compensation
Power Control	LPn / XSHUT Usage	Not Used	N/A	Hardware shutdown strategy
	Autonomous Low Power (ALP) Enabled	Yes	N/A	Sensor self-triggering
	Interrupt Enabled	No	Yes	Used to wake MCU
	Inter-measurement Period	Default	Default	Time between measurements
Microcontroller Interaction	ESP32 Light Sleep Between Frames	No	Yes	Reduces MCU duty cycle
	Wake Source	Timer	Sensor Interrupt	Timer vs Sensor Interrupt
	I2C Speed	Default (400 kHz - Fast Mode)	Default (400 kHz)	Can affect transaction time
Adaptive Logic (Firmware-Level)	Adaptive Frequency Scaling	Not Implemented	No	Yes/No
	Frequency Scaling Logic Basis	N/A	1 Hz (objects >300 cm from lift), 6 Hz (objects <300 cm from lift)	Distance trend / threshold based
	Threshold Distance for Mode Change	N/A	Not implemented for this system	e.g., 1.0 m
	Hysteresis Implemented	No	Yes	Prevents rapid toggling

	Stability Time Before Downscaling	N/A	Yes (4 seconds)	e.g., 5 sec stable
System Architecture	Continuous Ranging Used	No	No (only during testing)	For baseline comparison
	Hardware Shutdown Mode Used	No	No (possible future implementation)	XSHUT implementation
	Multi-Sensor Coordination	N/A	In Progress	If scaling later
Final Selected Architecture	Overall Operating Strategy	Baseline Default	ALP + Light Sleep + Adaptive 1–6 Hz Scaling + 4x4 Resolution + 10 VHV Count + CLOSEST target order	Example: “ALP + Light Sleep + Adaptive 1–5 Hz Scaling”